RIGA TECHNICAL UNIVERSITY
Faculty of Computer Science, Information Technology and Energy
Institute of Applied Computer Systems

Aum Sohoni
Student of the Academic Bachelor Level Study Program "Computer Systems"
Student ID No 241ADB009

RECIPROCAL HUMAN-AI JOB MATCHING PLATFORM FOR SKILLED TRADES

BACHELOR THESIS
6th Semester Report

Scientific adviser
Lecturer Janis Amolins

RIGA 2026

ABSTRACT

RECIPROCAL MATCHING, ACTIVE LEARNING, SKILLED-TRADES RECRUITMENT, HYBRID HUMAN-AI SYSTEMS, ALGORITHMIC GOVERNANCE, EU AI ACT, INFORMATION RETRIEVAL EVALUATION

This report describes the research, design, implementation and early testing of MeisterMatch, a mobile-first reciprocal human-AI recruitment platform for skilled trades (electricians, plumbers, welders, carpenters and HVAC technicians) in Latvia and the Baltic region (Hazans, 2019; European Labour Authority, 2026). It addresses the mismatch between text-dense job boards and informal hiring: personal referrals remain a common and effective way to fill vacancies in developed labour markets (Hoffman, 2017), but such matches leave neither side with an inspectable, data-backed record.

MeisterMatch combines a gesture-driven swipe-card interface with a readable multi-factor scoring engine whose factor weights can be adjusted from recruiter feedback. To align with Regulation (EU) 2024/1689 (European Parliament & Council, 2024a) and Directive (EU) 2024/2831 (CMS Law, 2024), it keeps a person in the decision loop. Rankings list plain match reasons, and recruiter feedback adjusts the factor weights through uncertainty sampling (Amershi et al., 2019).

The prototype has a Python/FastAPI ranker, a Node.js/Express API and a React 18/Vite frontend. Chapter 4 verifies that all 13 API endpoints run, that types stay consistent, that passwords are stored as salted scrypt hashes and that session tokens expire, and that the seed data (30 workers, 20 jobs) flows end to end. Chapter 5 reports an executed controlled experiment, over 80 replicate markets at three market sizes, that compares the active-learning ranker with the fixed-weight baseline on Precision@K and NDCG; it finds a clear gain on the larger markets and no distinguishable gain on the smallest. Voice onboarding, calendar scheduling and messaging bots are out of scope.

Report parameters: bachelor thesis progress report; approximately 40 pages including the title page, the list of references and the appendices; 11 tables, 3 figures, 36 references.

Keywords: Reciprocal Matching, Active Learning, Skilled-Trades Recruitment, Hybrid Human-AI Systems, Algorithmic Governance, EU AI Act, Information Retrieval Evaluation.

ANOTĀCIJA

Šajā bakalaura darba starpatskaitā aprakstīta MeisterMatch izstrāde — savstarpēja (reciprocal) cilvēka un mākslīgā intelekta darba meklēšanas platforma kvalificētiem amatniekiem Latvijā. Platforma apvieno žestu vadītu karšu saskarni ar caurskatāmu, uz noteikumiem balstītu daudzfaktoru vērtēšanas dzinēju, kurā cilvēks paliek lēmumu pieņemšanas ciklā. Atskaite ietver problēmas izklāstu, mērķi un uzdevumus, informācijas avotu analīzi, prototipa praktisko risinājumu un tā verifikācijas sākumu.
Atslēgvārdi: savstarpēja saderība, aktīvā apmācība, amatniecības profesijas, cilvēka un MI sadarbība, Eiropas Savienības MI akts.

TABLE OF CONTENTS

ABSTRACT....2
ANOTĀCIJA....3
TABLE OF CONTENTS....4
INTRODUCTION....6
1. LABOUR MARKET CHALLENGES AND THESIS OBJECTIVES....8
1.1. Statement of the Problem....8
1.2. Goal of the Thesis....9
1.3. Tasks of the Thesis....9
1.4. Scope Boundaries and De-Scoped Features....10
1.5. Justification of Topicality....10
1.6. Development Schedule....10
2. ANALYSIS OF INFORMATION SOURCES (ANALYTICAL PART)....12
2.1. Skilled Trades Recruitment Dynamics in Latvia and the Baltics....12
2.2. Hybrid Human-AI Architectures and Algorithmic Matching....13
2.3. Active-Learning Feedback Mechanisms....14
2.4. Mobile Gesture UX, Reciprocal Matching and Behavioural Dynamics....15
2.5. EU Legal Compliance, Data Privacy and Algorithmic Governance....15
2.6. Related Marketplaces for Skilled-Trades Work....16
2.7. Comparison of Existing Recruitment Platforms....17
2.8. Results of the Analytical Part....17
3. PRACTICAL SOLUTION (PRACTICAL PART)....19
3.1. System Architecture....19
3.2. Data Model....20
3.3. Matching Engine Implementation....21
3.4. Authentication and Session Model....22
3.5. Key Interaction Sequences....23
3.6. Seed Data and Cold-Start Rationale....24
3.7. User Interface Design and Interaction....24
3.8. Development Status and Results of the Practical Part....25
4. PRACTICAL SOLUTION VERIFICATION....26
4.1. Verification Approach....26
4.2. Build and Schema Verification....26

4.3. API Endpoint Verification 27
4.4. Frontend and End-to-End Integration Verification 27
4.5. Verification Results and Test Matrix 27
4.6. Security Review, Known Limitations and Planned Hardening 28
5. EMPIRICAL EVALUATION METHODOLOGY AND PRELIMINARY RESULTS 30
5.1. Information Retrieval Metrics 30
5.2. Baseline Comparison Protocol 30
5.3. Threats to Validity 32
5.4. Ethical Considerations 32
5.5. Reproducibility 32
5.6. Decision Criteria 32
5.7. Preliminary Controlled Experiment (This Stage) 33
RESULTS AND CONCLUSIONS 35
LIST OF REFERENCES 37
APPENDIX: SHARED DATA CONTRACT AND API SURFACE (EXTRACT) 40

INTRODUCTION

Latvia, and the Baltics more broadly, keep running short of vocational workers, and this has been the case for years. Skilled trades occupations, including certified electricians, plumbers, industrial welders, carpenters and HVAC technicians, are among the hardest roles to fill in construction, manufacturing and municipal infrastructure. Older job sites (such as CV-Online, CV Market and Visidarbi.lv) rely on lengthy text descriptions, complex filter matrices and asynchronous resume uploads. These conventional interfaces serve the skilled trades sector poorly, because trades workers rely on informal peer networks, prioritise immediate local availability and wage transparency, and show low engagement with desktop applicant-tracking systems.

AI screening tools promised to help by ranking CVs automatically. On their own, however, they are hard to inspect, they can repeat historical biases and they clash with emerging EU rules. While algorithmic screening can save time, autonomous decision-making brings risks of black-box opacity, reinforced historical bias and regulatory non-compliance (Rigotti & Fosch-Villaronga, 2024). Under Regulation (EU) 2024/1689 (the European Union Artificial Intelligence Act), AI applications in recruitment are designated as high-risk systems requiring human oversight, transparency and data governance.

This report addresses these challenges by designing, building and checking MeisterMatch, a reciprocal human-AI job-matching platform for skilled-trades recruitment in Latvia (Hazans, 2019; European Labour Authority, 2026).

The goal of the bachelor thesis is formulated as follows: to design, implement and verify a prototype of a reciprocal human-AI skilled-trades recruitment platform for the Latvian labour market, featuring a gesture-driven mobile interface, an interpretable multi-factor ranking engine, an active-learning feedback loop, and design consistent with EU human-oversight and transparency requirements for high-risk AI applications (European Parliament & Council, 2024a).

The tasks of the thesis are: (1) literature analysis, (2) hybrid architecture design, (3) architectural specification, (4) prototype implementation, (5) active-learning integration, (6) software verification and (7) evaluation methodology. Mutual interest is established strictly through dual consent, so contact details are unlocked only when both candidate and employer express interest. A stated trust limitation is that informal

referral trust is not modelled: “verified trade skills” currently means self-declared skills shown on the card, and referral or employer-endorsement features are future work. The swipe-UI friction argument rests on limited evidence (Iyengar & Lepper, 2000; Raghavendra & Chitra, 2025) and would need a short usability test with tradespeople to confirm.

The report is organised as follows. The Introduction sets out the topicality, the goal, the tasks and an overview of the chapters. Chapter 2 presents the analytical part: a review of information sources on the Baltic skilled-trades labour market, hybrid human-AI architectures, active-learning methodology, mobile gesture interaction, reciprocal matching and EU legal compliance, closing with the results of the analysis. Chapter 3 presents the practical part: the three-tier prototype, its data model, the matching engine, the user interface and the development status. Chapter 4 covers the verification of the practical solution that has already started. Chapter 5 sets out the empirical evaluation methodology and the results of a preliminary controlled run executed in this stage. The report closes with results and conclusions, a list of references and appendices.

The methodology follows a design-science approach. A structured literature review established the requirements and the constraints; a system architecture was then specified and implemented as a working prototype; software verification confirmed build integrity, schema consistency and API reliability; and an evaluation protocol was defined to benchmark the interpretable ranker and its active-learning extension, with a preliminary run executed and reported in Chapter 5.

1. LABOUR MARKET CHALLENGES AND THESIS OBJECTIVES

1.1. Statement of the Problem

The Latvian labour market has a clear supply-demand gap in vocational technical trades. Studies of Latvian emigration (Hazans, 2019) show that emigration, peaking after the 2008-2009 crisis, removed a large share of the working-age population from manual, craft and technical trades: between 2008 and 2017 about 260,000 people left the country, equivalent to 13.5% of the resident population in 2017 (OECD, 2019a). This outflow has contributed to persistent shortages in skilled trades that education and VET reforms have so far not offset (Cedefop, 2024).

According to official labour-market forecasts published by the Latvian Ministry of Economics (2020), Latvia will face a shortage of around 40,000 workers with secondary vocational qualifications in the medium term, alongside a persistent surplus of low-skilled workers. The EURES report of the European Labour Authority (2026) lists craft workers, building electricians, welders and plasterers among the widespread shortage occupations in Latvia; roofers and plumbers are not reported as shortage occupations for Latvia, although they are reported as shortages elsewhere in the EU. Traditional online portals only partly resolve this gap, because trades recruitment in Latvia still relies heavily on personal networks and referrals (Hoffman, 2017). Search frictions of this kind are precisely the inefficiency that electronic labour-market matching is meant to reduce (Autor, 2001), yet the design of most job boards is ill-suited to this segment.

The shortage is felt most acutely by small and medium enterprises, which dominate construction, manufacturing and property maintenance in Latvia and rarely have dedicated recruitment staff: vacancy postings are created manually, applications arrive asynchronously, and shortlisting relies on memory and informal reputation rather than structured data. On the worker side, certified tradespeople hold several active skills and move between engagements of varying length, so a browsable profile that states availability and an hourly rate is more useful than a date-stamped application. These observations drive the interface and ranker choices made in Chapter 3.

1.2. Goal of the Thesis

The primary goal of this bachelor thesis is to design, implement and verify a prototype of a reciprocal human-AI skilled-trades recruitment platform for the Latvian labour market featuring a mobile gesture-driven interface, an interpretable multi-factor ranking engine, an active-learning feedback loop, and design in line with EU human-oversight and transparency requirements for high-risk AI applications.

1.3. Tasks of the Thesis

To achieve the defined goal, seven specific academic and engineering tasks have been established:

1. Literature Analysis: analyse academic literature, official labour-market statistics (EU-LFS, Ministry of Economics) and EU legal frameworks regarding skilled-trades recruitment, hybrid human-AI architectures and active learning.
2. Hybrid Architecture Design: investigate hybrid human-AI architectural patterns (Dellermann et al., 2019) in order to design an operational human-in-the-loop governance mechanism complying with the EU AI Act.
3. Architectural Specification: formulate a multi-tier software architecture separating cross-lingual candidate representations, interpretable ranking logic and real-time state persistence.
4. Prototype Implementation: implement a functional web prototype comprising worker and employer card decks, swipe navigation, a rule-based matching engine and mutual-interest messaging.
5. Active-Learning Integration: implement an active-learning feedback loop using uncertainty sampling to refine recommendation weights from recruiter interactions, and demonstrate it in a controlled comparison against the fixed weights.
6. Software Verification: verify software build integrity, schema contracts and REST API reliability across a deterministic seed dataset of 30 workers and 20 job postings.
7. Evaluation Methodology: formulate an empirical evaluation methodology (Precision@K, Recall@K, task time) to benchmark the active ranker against a non-learning baseline; a preliminary run of this methodology over replicate markets is reported in Chapter 5.

1.4. Scope Boundaries and De-Scoped Features

A bachelor thesis cannot cover everything, so the scope was kept narrow. The scope covers only reciprocal matching, multi-factor interpretable ranking, active-learning feedback and baseline evaluation. Secondary feature extensions, specifically voice onboarding via speech recognition, calendar scheduling integration and automated Telegram notification bots, were excluded from the assessed prototype and identified strictly as future work. A further scope note: the original idea included urgent on-demand dispatch (a client needing a plumber immediately). This thesis instead covers longer-term reciprocal hiring through mutual swiping; urgent two-hour dispatch is explicitly out of scope.

1.5. Justification of Topicality

The relevance of this work comes from three points:

- Macroeconomic urgency: labour and skills shortages are a binding constraint on Latvia's economy: 90% of firms cited skilled-labour shortages as a barrier to long-term investment in 2025, and the skills-mismatch rate rose to 18.7% in 2024 (European Commission, 2026). The national priority projects outlined in the National Development Plan of Latvia 2021-2027 (Cross-Sectoral Coordination Centre, 2020), including housing renovation and the energy transition, are directly bottlenecked by vocational trade shortages.
- Regulatory compliance: Regulation (EU) 2024/1689 (European Parliament & Council, 2024a) designates AI systems used in recruitment as high-risk and mandates human oversight, transparency and data governance. Developing an interpretable, human-in-the-loop platform provides a practical model for regulatory compliance (Rigotti & Fosch-Villaronga, 2024).
- User-experience alignment: gesture-driven mobile interfaces lower friction for blue-collar trade workers compared with desktop applicant-tracking forms.

1.6. Development Schedule

The 19-week schedule for the bachelor thesis project is summarised in Table 1.1.

Table 1.1

Bachelor Thesis Development Schedule

Activity	Deadline
Collect and survey literature sources	Week 3
Analyse information sources and draft the analytical chapter	Week 6
Submit the progress report (this document)	Week 9
Design the system architecture and data model	Week 10
Implement the matching engine and the active-learning loop	Week 11
Pre-defence	Week 12
Conduct the evaluation and write the results chapter	Week 15
Write the ethical implications and conclusions	Week 16
Bachelor thesis submission	Week 17 (Tuesday)
Bachelor thesis defence	Week 19

2. ANALYSIS OF INFORMATION SOURCES (ANALYTICAL PART)

The analysis of information sources draws on peer-reviewed academic literature, official labour-market statistics, industry reports, technical documentation and EU policy documents. Sources were identified through Google Scholar, IEEE Xplore, the ACM Digital Library, Crossref and official statistical and regulatory portals. The sections below summarise the findings by thematic area: skilled-trades recruitment dynamics in Latvia and the Baltics, hybrid human-AI architectures for matching, active-learning feedback mechanisms, mobile gesture interaction and reciprocal matching, and EU legal compliance and algorithmic governance. The chapter closes with a comparison of existing platforms and the conclusions drawn from the analysis.

2.1. Skilled Trades Recruitment Dynamics in Latvia and the Baltics

The local picture is based on EU Labour Force Survey and regional labour-market studies (Hazans, 2019; OECD, 2019a; OECD, 2019b). Professional white-collar hiring relies on commercial job portals (CV-Online, CV Market), whereas skilled-trades recruitment leans on informal networks, personal recommendations and local classified portals (for example ss.lv / ss.com). OECD evaluations find that registered vacancies account for less than 20% of all hires (OECD, 2019b), and referral hiring lowers recruiting costs and turnover in developed markets, though it can reinforce homophily (Hoffman, 2017). In Latvian practice (European Labour Authority, 2026), informal networks dominate trades recruitment because they provide peer trust and verification of practical craftsmanship, which static CV text fails to communicate.

The mismatch is structural rather than cyclical. The post-2008 emigration wave removed experienced manual workers faster than the domestic education system could replace them, and vocational enrolment never fully recovered (Hazans, 2019; OECD, 2019a). Because trades work is location-bound and often requires Latvian or Russian language competency, international replacement talent is only a partial answer, which strengthens the case for better domestic matching rather than additional supply alone (European Labour Authority, 2026). The economic logic of lowering search frictions in such a market is well established: electronic matching

raises the efficiency of vacancy-candidate contact even when the underlying skills gap persists (Autor, 2001).

At the EU level the signals are consistent: the European Labour Authority's shortage report confirms that craft workers, building electricians, welders and plasterers are among the occupations with recognised shortages in Latvia (European Labour Authority, 2026). Because EURES covers only formally reported vacancies, these counts under-state informal hiring, so combining EURES data with labour-force-survey evidence and the referral literature gives a more complete picture (Hazans, 2019; OECD, 2019b). Two implications follow: the scarcity of trades labour is a matching problem as much as a supply problem, so improving contact efficiency is an immediately actionable contribution (Autor, 2001); and a tool for this market must expect many economically active candidates who are not browsing vacancy boards, which motivates the push-oriented card deck proposed here.

2.2. Hybrid Human-AI Architectures and Algorithmic Matching

The introduction of algorithmic recommendation engines into labour marketplaces affects matching efficiency. In a large-scale field experiment on a digital labour market, Horton (2017) showed that providing employers with algorithmically generated candidate recommendations increased job fill rates for technical vacancies by 20% compared with control groups, without crowding out organic non-recommended applicants. Because the experiment ran in an online freelance market rather than in Latvian skilled-trades recruitment, external validity is limited: such recommendations should be read as complementing, not substituting for, costly manual screening.

Fully automated decision systems, however, introduce risks of systematic bias and opacity. Statistical discrimination can arise even from accurate proxy variables, so disparate impact is a structural property of data-driven screening rather than an isolated bug (Barocas & Selbst, 2016). In digital work settings more broadly, algorithmic control reshapes the relationship between platforms and workers and is contested by the people subject to it (Kellogg et al., 2020). To mitigate these risks, MeisterMatch adopts the hybrid-intelligence framework formulated by Dellermann et al. (2019). Hybrid intelligence combines complementary human cognitive abilities (contextual judgment, ethics, domain intuition) with machine capabilities (rapid pattern recognition, multi-criteria scoring). By embedding recruiters directly in the

decision loop, the system functions as a supportive recommender rather than an autonomous decision-maker. These interaction principles follow Amershi et al. (2019).

Within the hybrid-intelligence view, the division of labour must be explicit. The machine contributes consistency: it applies the same scoring rules to every pair and quantifies each factor. The human contributes context: whether an employer replies, whether a candidate is reachable, whether a borderline score is worth an interview. This keeps the system defensible under the EU AI Act, because the recommendation is an input to a human decision rather than the decision itself (European Parliament & Council, 2024a). Following the interaction guidelines of Amershi et al. (2019), each deck card renders the factor bars and plain-language reasons, so a recruiter can agree with, override or refine a ranking - the first step toward the active-learning feedback described in Section 2.3 and evaluated in Section 5.7.

2.3. Active-Learning Feedback Mechanisms

Active learning here means the following. The build records recruiter likes and passes (like = 1, pass = 0) and fits the ranker's factor weights to that feedback. The learning layer is a logistic regression on the five factor scores (skill overlap, district match, availability match, pay fit and experience): a weighted sum, squashed by the logistic function, supplies a calibrated probability, and each feedback batch performs one ridge-regularised update of the weights by iteratively reweighted least squares. Sampling then presents unseen pairs whose predicted probability is closest to 0.50 (maximum uncertainty). This follows the active-learning paradigm in which the learner queries the most informative points (Settles, 2012).

Updates are evaluated on labels the learner never trains on. Each replicate holds out an evaluation share of the market pairs and a validation share of the feedback; the evaluation share scores Precision@K and NDCG, and the validation share triggers a rollback that reverts an update whenever validation NDCG drops more than a fixed margin below its best value. Ridge regularisation toward the starting weights and a bounded weight corridor limit how far one batch can move a factor, and the learner only adjusts factor weights, never the trade filter or the pay and availability rules that encode hard requirements. Active preference elicitation reaches higher accuracy from fewer interactions than passive rating collection (Elahi et al., 2016; Rubens et al., 2015), while multilingual models such as XLM-R (Conneau et al., 2020; Hu et al.,

2020) remain a candidate feature extractor for the final stage only and are not connected to the current ranker.

2.4. Mobile Gesture UX, Reciprocal Matching and Behavioural Dynamics

Gesture-driven swipe interactions for job matching were operationalised by Raghavendra & Chitra (2025) in SwipeRight, where combining swiping with real-time state synchronisation shortened screening cycles and increased engagement. Reciprocal models score candidate-job pairs from both sides of the market (Yao et al., 2023), and reciprocal recommender systems more generally model mutual rather than one-sided preference (Palomares et al., 2021). Palomares et al. distinguish score-based methods, which compute a utility for each side and combine them, from collaborative and content-based methods. MeisterMatch belongs to the score-based family: jobs-for-worker and workers-for-job scoring are computed independently, and mutual interest comes from the independent like signals rather than a single combined utility, so neither side's score reveals the other's. The platform is a two-sided market with the network effects and cold-start dynamics that platform economics describes (Rochet & Tirole, 2003; Schein et al., 2002).

On the behavioural side, large, undifferentiated choice sets can undermine motivation (Iyengar & Lepper, 2000). Waleson (2022) compared fast card-swiping with slow browsing in a mock dating app and found no significant wellbeing effect, so whether curated, pre-ranked decks prevent choice overload in job search remains a hypothesis. Gamified mechanics such as instant match feedback are widely used to motivate candidates (Obaid et al., 2020). To allow early testing despite network sparsity, the prototype uses deterministic synthetic seed profiles (30 workers, 20 jobs); this follows the general bootstrapping pattern discussed by Jobtech Alliance (2026), not their recommendation of these exact numbers.

2.5. EU Legal Compliance, Data Privacy and Algorithmic Governance

Because EU rules apply to hiring AI, the design includes the following controls:

- EU AI Act (European Parliament & Council, 2024a): under Regulation (EU) 2024/1689, AI systems used in recruitment and employment are classified

as high-risk (Annex III). The law mandates risk management, data governance, transparency and human oversight.

- Platform Work Directive (European Parliament & Council, 2024b; CMS Law, 2024): applies to digital labour platforms and to persons performing platform work; its relevance here depends on whether matchmaking counts as algorithmic management, which is an open legal question. Its limits on processing sensitive personal data are narrower than a general ban. Related GDPR Art. 22 safeguards on solely automated decisions are also considered in the design.
- GDPR compliance: in a systematic literature review, Amaral Cejas et al. (2026) note that mobile apps frequently violate data-subject rights (access, erasure, consent withdrawal). MeisterMatch implements strict data minimisation and user-controlled profile visibility.

Human review is not claimed to guarantee fairness, because human recruiters also possess cognitive biases (Köchling & Wehner, 2020; Raghavan et al., 2020). Human review provides a mechanism to inspect, contest and override recommendations; it does not remove bias on its own.

2.6. Related Marketplaces for Skilled-Trades Work

A cross-sectional look at existing consumer-facing marketplaces for trades work positions MeisterMatch relative to commercial practice. Platforms such as Hipages (Australia), MyBuilder (United Kingdom) and Thumbtack (United States) are service-sourcing marketplaces: a household posts a job, vetted professionals respond with quotes, and the platform mediates trust through reviews and insurance. From a matching-theory viewpoint they are one-sided response markets rather than reciprocal markets, because the household selects among responders while the professional's decision problem (which jobs to pursue, at what price) stays implicit. They are also quote-based and transaction-heavy, which suits one-off household repairs less well than the continuous or project-based staffing relationships that construction and municipal employers in Latvia need (Hazans, 2019).

Two-sided recruitment for trade labour, by contrast, resembles the graduate-referral matching of Yao et al. (2023), in which both sides swipe and a match requires mutual interest, or the swipe mechanics of Raghavendra & Chitra (2025). MeisterMatch occupies this category: a hiring tool for employers who already need

electricians and welders, not a lead-generation board. The unit of value is a staffing relationship rather than a quotation, so the platform emphasises stated availability, hourly rates and district proximity. To the author's knowledge, no current platform combines all four properties investigated here - reciprocal dual-consent matching, an interpretable multi-factor ranker, an active-learning feedback loop and alignment with EU high-risk AI obligations - which is the gap this work aims to close.

2.7. Comparison of Existing Recruitment Platforms

To position the prototype, the reviewed platforms and tools are compared in Table 2.1. The comparison is qualitative and was synthesised by the author from the sources reviewed in this chapter rather than from controlled measurement.

Table 2.1

Qualitative Comparison of Existing Recruitment Platforms and MeisterMatch

Platform / tool	Primary sector	Interface	Matching	Human oversight
CV-Online, CV Market, Visidarbi.lv	General / white-collar	Web forms, keyword filters	Manual search	Recruiter decides
ss.lv / ss.com	Classifieds	Listing board	Manual browsing	None
LinkedIn	Professional / knowledge work	Profile + feed	Algorithmic ranking	Recruiter decides
Enterprise AI screening tools	Large employers	ATS-integrated	Automated ranking	Limited
MeisterMatch (this thesis)	Skilled trades	Mobile swipe cards	Reciprocal + interpretable ranker	Human-in-the-loop

2.8. Results of the Analytical Part

The analytical literature review produced four findings that shaped the system design:

1. Informal local dynamics: skilled-trades recruitment in Latvia leans heavily on informal networks and personal referrals (Hoffman, 2017), which is why this thesis explores a mobile-first, low-friction interface as one possible way to reduce interaction barriers.
2. Recommender liquidity: algorithmic recommendations increase job fill rates by 20% for technical vacancies in online labour markets (Horton, 2017),

but must preserve human oversight under the EU AI Act (European Parliament & Council, 2024a).

3. Active feedback efficiency: active learning with uncertainty sampling optimises feature weights under sparse feedback constraints (Settles, 2012; Elahi et al., 2016).
4. Gesture UX benefits: choice overload is a documented risk when options are numerous and undifferentiated (Iyengar & Lepper, 2000); whether pre-ranked card decks prevent it in job search is a hypothesis to be tested (Waleson, 2022 found no wellbeing effect of swiping speed).

The analysis also confirmed a design implication from platform economics: as a two-sided market, MeisterMatch must solve a cold-start problem on both sides, which is why the prototype seeds both worker and job supply with a deterministic dataset (Rochet & Tirole, 2003; Schein et al., 2002).

The findings of this chapter translate directly into a requirement set for the prototype. Table 2.2 organises the requirements into functional ones (what the system must do) and non-functional ones (how it must behave), each anchored in the section of the analysis that motivated it.

Table 2.2

Functional and Non-Functional Requirements Derived from the Analytical Part

Type	Requirement	Origin in the analysis
Functional	Reciprocal dual-consent matching	Referral and two-sided market findings (Sec. 2.4, 2.6)
Functional	Interpretable per-factor reasons on every recommendation	Transparency and hybrid-intelligence findings (Sec. 2.2, 2.5)
Functional	Hard trade filtering before scoring	Skilled-trades market structure (Sec. 2.1)
Functional	Swipe-derived like/pass feedback recorded as labels	Active-learning design (Sec. 2.3)
Non-functional	Mobile-first gesture interface	User-experience findings (Sec. 2.4)
Non-functional	Human-in-the-loop oversight; no autonomous decisions	EU AI Act compliance (Sec. 2.5)
Non-functional	Data minimisation and mutual-match-only visibility	GDPR review (Sec. 2.5)
Non-functional	Deterministic, reproducible seed market for evaluation	Cold-start and evaluation needs (Sec. 2.7, 2.8)

3. PRACTICAL SOLUTION (PRACTICAL PART)

3.1. System Architecture

MeisterMatch uses a separated three-tier design for easier maintenance, type safety and real-time state synchronisation. Fig. 3.1 illustrates the structural components and data flow across the system.

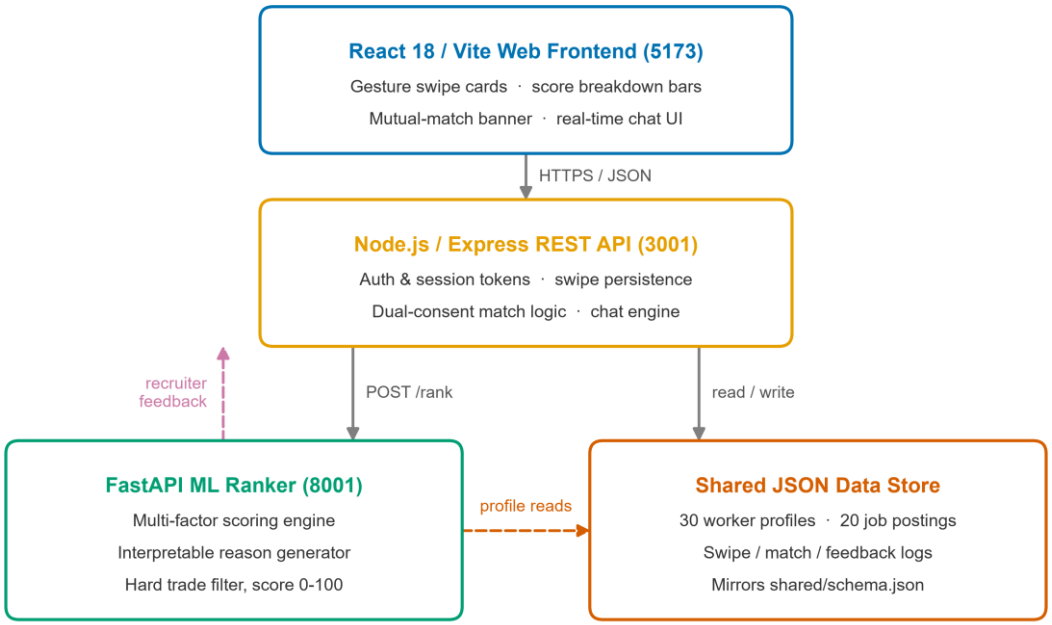


Fig. 3.1. System Architecture and Active-Learning Feedback Data Flow

The frontend (React 18 / Vite, port 5173) renders swipe-based job and worker cards, sends like/pass actions and displays match results. The API layer (Node.js / Express, port 3001) handles authentication, swipe recording, dual-consent match creation and chat messaging. The ML ranker (Python / FastAPI, port 8001) receives candidate-job pairs and returns scored, ranked results with interpretable reasons. A shared JSON data store holds the profiles, swipes, matches and feedback and is the runtime source of truth. Table 3.3 summarises the three services and the shared contract.

Table 3.3

Services, Technologies and Ports of the MeisterMatch Prototype

Service	Technology	Port	Responsibility
ML ranker	Python / FastAPI (uvicorn)	8001	Multi-factor scoring and interpretable reasons
Application API	Node.js / Express	3001	Auth, swipes, matches, chat and deck proxies
Web client	React 18 / Vite	5173	Card decks, score bars and chat UI

Shared contract	schema.json (mirrored in contract.ts, models.py)	-	Single source of truth for types

The services communicate over plain HTTP with JSON. The API proxies ranking requests to the ranker for the two deck routes (jobs-for-worker and workers-for-job), so the frontend never talks to the Python service directly. This keeps the internal interface surface small and testable: every inter-service conversation is a single documented JSON request to /rank.

3.2. Data Model

The data contract is shared between all three services through a single JSON-Schema file (shared/schema.json) that is mirrored in TypeScript (shared/contract.ts) and Python (shared/models.py). This guarantees type consistency across the TypeScript frontend, the TypeScript API and the Python ML service. The principal entities are summarised in Table 3.1.

The contract is strict about identifiers and enumerations. Worker and job identifiers follow the patterns w### and j###; the trade field is restricted to electrician, plumber, welder, carpenter and hvac; languages are limited to lv, ru and en; availability and job start share the same three-state enumeration (immediate, one-to-two-weeks or a specific date); and jobs additionally declare a season (summer, winter or all-year) and carry a Latvian description field (description_lv). The fields consumed by the ranker are the enumerated ones plus the hourly rate band.

Table 3.1

Principal Data-Model Entities in the MeisterMatch Contract

Entity	Key fields	Purpose
Worker	id, trade, skills[], experience_years, district, languages[], availability, hourly_rate_eur	One skilled-trades professional profile
Job	id, employer, title, trade, required_skills[], district, pay_min/max, start, season	One employer vacancy
Swipe	actor_type, actor_id, target_type, target_id, direction (like/pass)	A single expression of interest
Match	id, worker_id, job_id	Created when both sides like each other
Feedback	worker_id, job_id, label (relevant/irrelevant), by	Label for the active-learning loop
User, ChatMessage	username, password_hash / text, by, time	Authentication and post-match messaging

3.3. Matching Engine Implementation

The matching engine (ml/ranker.py) calculates candidate-job relevance using a transparent point-based scoring function over five factors, with a hard trade filter that removes non-matching trades before scoring. The five factors are skill overlap, same-district geography, availability, pay compatibility and experience. Fig. 3.2 shows the maximum points each factor can contribute.

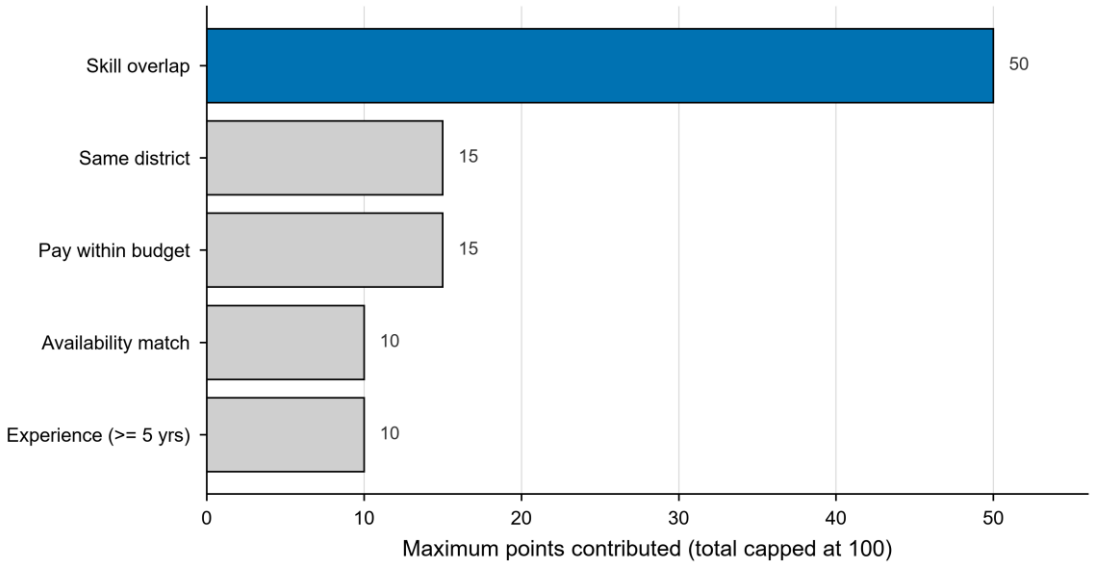


Fig. 3.2. Maximum Points Contributed by Each Interpretable Scoring Factor

Points are awarded per factor: skill contributes up to 50 points (50 times the number of matched required skills divided by the number of required skills; 25 points if the job lists no required skills); same-district geography adds 15; availability adds 10 for an exact availability match or 4 if either side is immediate; pay within the employer's budget adds 15 (or 8 if the rate is within 10%); and experience adds 10 for 5 or more years, 5 for at least 1 year and 2 otherwise. The total is capped at 100. There is no language factor in the current build, because job posts do not carry required languages; language profiles exist on worker records only. A limitation is that the geographic factor currently covers ten seeded Riga districts (Centrs, Purvciems, Agenskalns, Imanta, Jugla, Kengarags, Plavnieki, Sarkandaugava, Teika, Ziepniekkalns); Latgale, Kurzeme and Vidzeme are not yet modelled, which limits the peripheral-region motivation and is left for the remaining work.

The geographic factor is a deliberate simplification. Proximity is a binary same-district signal: identical districts add 15 points and any different-district pair adds none, so a pair across two adjacent districts scores exactly like a pair on opposite

sides of the city. A finer proximity model would act on a limited signal in the current market, because the executed experiment of Section 5.7 recovers a district sensitivity in the lower half of the five factors, below availability and pay; such a model is left to future work.

Worked example. For an electrician in Purvciems with 3 years of experience, availability in one-to-two weeks and a rate of 20 EUR/h, matched against a Centrs electrician job requiring Wiring, HVAC and PLC, starting immediately and paying 16-21 EUR/h, the factor points are: skill overlap 1/3 × 50 = 17, same district 0, availability +4, pay +15 and experience +5, giving a total of 41. This illustrative calculation follows the documented rules exactly. Table 3.2 shows the same calculation row by row.

Table 3.2
Worked Example of the Interpretable Scoring Rules for One Candidate-Job Pair

Factor	Rule applied	Max	Scored	Reason shown on the card
Skill overlap	50 × matched / required (25 if none)	50	17	Partial skills match (Wiring)
Same district	+15 if worker and job share a district	15	0	No same-district bonus
Availability	+10 if equal, +4 if either side is immediate	10	4	Availability partly overlaps job start
Pay fit	+15 within budget, +8 close to budget	15	15	Pays the candidate's rate (up to 21 EUR/h)
Experience	+10 for 5+ years, +5 for 1+, +2 otherwise	10	5	Some experience (3 yrs)
Total	Sum of factor points, capped at 100	100	41	Composite score

The engine outputs explicit, human-readable match reasons for every scored pair (for example "Partial skills match (Wiring)" or "Located in same district (Centrs)"), so the result stays open to inspection. Weights start as fixed constants in the current build; the active-learning layer described in Section 2.3 and evaluated in Section 5.7 adjusts these weights from recruiter feedback.

3.4. Authentication and Session Model

Authentication follows a token-based design with hashed credentials and expiring sessions. The API exposes four auth routes: register, login, logout and a current-user route. On register, the username and password are validated (the username must be at least three characters), the password is stored as a salted scrypt

hash (16-byte salt, 32-byte key, via node:crypto) rather than in plaintext, a user record is appended to the shared store and a bearer token is returned to the client. Login verifies the submitted password against the stored hash and issues a 192-bit random token with a 24-hour expiry; tokens are held in an in-memory session table, and expired tokens are rejected and pruned on use. The current-user route returns the current username when the request carries a valid, unexpired token. Logout deletes the token from the session table. Every route except /health, register and login requires the Authorization: Bearer <token> header, and requests without a valid token receive a 401 response. The web client stores the token obtained at login and attaches it to all subsequent API calls. Records created by earlier builds that stored plaintext passwords are upgraded to a scrypt hash on first successful login. The remaining security limitations of this prototype are reviewed in Chapter 4.

3.5. Key Interaction Sequences

The core usage flows are best described as short sequences of API calls rather than as screens:

1. Worker browsing: the worker logs in, requests the ranked job deck (GET /api/deck/worker/w001) and swipes right or left on each card; every swipe records a like or a pass for that worker-job pair (POST /api/swipes).
2. Employer browsing: the employer logs in, requests the ranked worker deck (GET /api/deck/job/j001) and evaluates candidate cards against the score bars and reason lists.
3. Mutual match: when the same worker-job pair has both a worker like and an employer like and no match exists yet, the swipe handler creates a Match record and returns it in the response, so the interface can immediately show the match banner and unlock the chat for both sides.
4. Post-match messaging: with the match id, either side lists (GET /api/chats/:matchId) or sends (POST /api/chats/:matchId) messages; each message carries the sender, the text and a server timestamp.

The dual-consent rule is enforced in exactly one place, the swipe handler, which checks that both like records exist before creating a match and refuses to duplicate an existing match. Contact details are not part of either profile type, so no private information changes hands before mutual interest has been established.

3.6. Seed Data and Cold-Start Rationale

Two-sided markets face a chicken-and-egg problem at launch: employers see little value without candidates, and candidates see little value without jobs (Rochet & Tirole, 2003; Schein et al., 2002). The prototype addresses this for development and evaluation with a deterministic, synthetic seed set rather than scraped or crowd-sourced data. The seed contains 30 worker profiles and 20 job postings spread across the five trades and the ten Riga districts used by the prototype, with realistic ranges for experience (a few years up to over twenty), hourly rates and availability states. Worker profiles include a trade, a skill list, experience, a district, language tags (lv, ru, en), availability and an hourly rate. Job postings include an employer, a title, required skills, a district, a pay band, a start state and a season. Because the seed is deterministic and versioned, every development run and every evaluation run observes the same market state, which makes the ranker output reproducible. The choice of 30 workers and 20 jobs balances breadth and audibility: the trades and districts are each well represented and the pool is small enough to inspect profile by profile. A market this small caps how much feedback a run can collect, so the controlled experiment of Chapter 5 does not rely on the pinned seed alone; it re-generates the market structure under many replicate seeds at three sizes and reports intervals across markets rather than single-seed point estimates.

3.7. User Interface Design and Interaction

The web frontend provides a card-swiping interface. Workers view pre-ranked job cards displaying employer details, hourly rates, required skills and score breakdown bars. Employers view candidate cards displaying declared trade skills, years of experience and proximity. Swiping right records a like; if both parties express mutual interest, the API instantly creates a match state and unlocks direct messaging. The interface uses a dark theme with glassmorphism elements, score bars and reason lists for each recommendation. Authentication is implemented with token-based sessions: users register or log in, receive a bearer token and include it in subsequent API requests.

The interface is intentionally minimal. The dark theme reduces glare in the low-light conditions in which tradespeople often plan their day, the score bars encode the interpretable factors in rank order, and the reason lists restate the ranker's plain-

language output. The layout is a single-column deck with a thumb-reachable action zone, following the mobile-first constraint established earlier.

3.8. Development Status and Results of the Practical Part

The current prototype is implemented and verified to the following extent: a rule-based multi-factor ranker, swipe decks, dual-consent matching, chat, scrypt password hashing with expiring sessions, and seed data of 30 workers and 20 jobs. The active-learning weight updater with uncertainty sampling and the controlled baseline comparison have also been implemented and executed, and are reported in Chapter 5. Future work beyond this thesis includes voice onboarding, calendar scheduling, Telegram notifications and a travel-time-aware proximity factor. By the end of the current stage the prototype included:

1. schema contracts: a shared data contract and JSON-Schema verification (shared/schema.json);
2. seed dataset: a deterministic seed dataset of 30 worker profiles and 20 job postings;
3. ML ranker service: a FastAPI ranker with multi-factor scoring and interpretable reasons;
4. REST API backend: an Express API handling auth, swipes, dual-consent match logic and chat;
5. React web app: a React/Vite frontend delivering gesture cards, score visualisation and chat UI.

The three services are launched together by start-all.ps1, which starts the Python ranker on port 8001 and then starts the web development server on port 5173 together with the API, opening the browser automatically. The Express API also serves the built frontend from web/dist when a production build exists, so the end-to-end demonstration runs identically from either the development or the production build.

4. PRACTICAL SOLUTION VERIFICATION

4.1. Verification Approach

Verification in this chapter only checks that the software runs (builds, schemas, API routes and end-to-end startup). It does not measure ranking quality. Ranking quality against a non-learning baseline (Precision@K, Recall@K, match success and task time) is the separate evaluation reported in Chapter 5. Verification was performed through automated build checks, API endpoint testing and manual inspection, with the goal of confirming that each service starts correctly, that the API endpoints respond with valid data and that the frontend renders without errors.

Evidence for every check listed in this chapter was captured as console output during the session in which the services were started with start-all.ps1 and exercised with curl and the browser. The word “verified” therefore means observed in this environment, not guaranteed by a test suite; the automated suite is part of the planned hardening described at the end of the chapter.

4.2. Build and Schema Verification

- TypeScript typechecking: ran “npm run typecheck” in the api/ and web/ directories without errors.
- Frontend asset build: ran “npm run build” in web/, producing production files in web/dist/.
- Schema consistency: confirmed that shared/schema.json correctly generates the TypeScript interfaces (contract.ts) and the Python Pydantic models (models.py).

Table 4.3 lists each check together with the command used to run it and the outcome observed during the verification session. The environment is a Windows development machine running the services through PowerShell: the ranker runs from a local Python virtual environment under uvicorn, the API and web client run through npm scripts, and start-all.ps1 orchestrates all three.

Table 4.3

Verification Checks Performed in This Stage

Check	Command	Observed outcome
TypeScript typecheck (api/)	npm run typecheck (api/)	No errors
TypeScript typecheck (web/)	npm run typecheck (web/)	No errors
Frontend production build	npm run build (web/)	Bundle written to

		web/dist/
Schema mirror consistency	contract.ts and models.py generated from shared/schema.json	Field names and enumeration values identical
Ranker startup	Start ranker on port 8001 (uvicorn)	Service up on :8001
API startup	npm run dev (api/)	Service up on :3001
End-to-end startup	start-all.ps1	All three services run; browser opens

4.3. API Endpoint Verification

The 13 core REST API routes were verified using automated curl scripts. Table 4.1 summarises the verification results.

Table 4.1

Verified REST API Endpoints and Functional Status

#	Method and route	Function
1	GET /health	Service liveness probe
2	POST /api/auth/register	Create a user and return a token
3	POST /api/auth/login	Authenticate and return a token
4	GET current-user route	Return the authenticated user
5	POST /api/auth/logout	Invalidate the session token
6	GET /api/workers	Return all 30 worker profiles
7	GET /api/jobs	Return all 20 job postings
8	GET /api/deck/worker/:id	Ranked job cards for one worker
9	GET /api/deck/job/:id	Ranked worker cards for one job
10	POST /api/swipes	Record a swipe and create a mutual match
11	GET /api/matches	Return active matches
12	GET /api/chats/:matchId	Return messages for a match
13	POST /api/chats/:matchId	Send a message for a match

4.4. Frontend and End-to-End Integration Verification

End-to-end orchestration was verified using the start-all.ps1 script, which started the FastAPI ranker (port 8001), the Express API (port 3001) and the Vite frontend (port 5173). Verification confirmed session persistence, card rendering, real-time mutual-match banner triggers upon reciprocal swiping, and instant chat message delivery. The browser opens automatically at http://127.0.0.1:5173.

4.5. Verification Results and Test Matrix

The checks performed so far and the planned hardening cases are summarised in Table 4.2.

Table 4.2

Verification Test Matrix (Completed and Planned)

Area	Check performed	Result	Remaining (planned)
Build	Typecheck api/ and web/	Pass	CI gate
Build	Production build of web/	Pass	Automated regression
Schema	contract.ts and models.py from schema.json	Consistent	Schema change tests
API (auth, incl. negative)	13 routes; 401 on missing, wrong or expired token; 409 on duplicate	Pass	400 invalid-body and invalid-match cases
End-to-end	start-all.ps1 brings up all services	Pass	pytest + supertest suite
Ranker	Deterministic on the seed set	Observed	Unit tests, empty-skill edge cases

Interpretation of the verification evidence. The accumulated checks support the claim, restated in the thesis goal, that a working prototype exists and behaves as specified by the schema. The two build checks pass without type errors, so the shared contract is respected by both the TypeScript client stack and the Python ranking service, and because contract.ts and models.py are both generated from shared/schema.json, a field added in one place cannot silently diverge from the other. The auth negative cases, the deterministic ranker behaviour on the seed set, and the end-to-end orchestration together demonstrate the full loop that the evaluation of Chapter 5 exercises: register, log in, receive a ranked deck, swipe, and observe mutual match and chat. What the evidence does not establish is any claim about match quality or user satisfaction; those are delegated to the controlled comparison of Chapter 5.

4.6. Security Review, Known Limitations and Planned Hardening

Security review of the current build. The two items flagged in review, plaintext password storage and missing session expiry, are now handled: passwords are persisted as salted scrypt hashes and session tokens carry a 24-hour expiry enforced on every authenticated request (Section 3.4). The remaining items stay on the hardening list: the API enables any origin, which must be restricted to the deployed frontend origin, and the shared JSON store has no concurrency control, so simultaneous writes could race. Neither affects the local evaluation.

The verification is a started activity, not a complete test cycle. Beyond the happy-path curl calls, the authentication negative cases were executed in this stage:

401 without a token, 401 for a wrong password, 401 for an expired token, and 409 for duplicate registration. Planned hardening includes invalid-body (400) and invalid-match cases, ranker unit tests, and an automated pytest and supertest suite. Known limits: no store concurrency control; an open CORS policy; and mutual-match-only profile visibility.

5. EMPIRICAL EVALUATION METHODOLOGY AND PRELIMINARY RESULTS

Chapter 4 covered software checks (verification). Chapter 5 is the separate evaluation: the protocol for testing whether the active ranker beats a fixed-weight baseline, and the results of a preliminary run of that protocol, executed in this stage on synthetic markets.

5.1. Information Retrieval Metrics

The evaluation framework uses four metrics:

1. Precision@K: the share of relevant candidates within the top K recommended cards (K = 3, 5, 10).
2. Recall@K: the share of total relevant candidate profiles successfully presented in the top card decks.
3. Match success rate: the percentage of candidate-job swipes that result in mutual-interest matches.
4. Recruiter task completion time: the time required, in seconds, for a recruiter to shortlist three qualified candidates.

Table 5.1 summarises the metric definitions and the way each value will be reported.

Table 5.1

Evaluation Metric Definitions used in the Study

Metric	Definition	Reported value
Precision@K	Share of relevant items within the top K = 3, 5, 10 recommendations	Mean over random seeds
Recall@K	Share of all relevant items shown in the top K decks	Mean over random seeds
Match success rate	Share of liked candidate-job pairs that become mutual matches	Percentage
Task completion time	Seconds needed to shortlist three qualified candidates	Median and interquartile range
NDCG@K	Rank-aware gain over truncation K	Mean over random seeds
MRR	Inverse rank of the first relevant recommendation	Mean over random seeds

5.2. Baseline Comparison Protocol

The active-learning ranker will be compared against a static fixed-weight baseline at feedback checkpoints (T = 0, 10, 25, 50, 75, 100 and 150 labelled pairs),

stated as H0 (no difference) versus H1 (active learning improves Precision@K and reduces task time). Ground-truth relevance labels are assigned by fixed rules documented in the test protocol (skill overlap ≥ 2 and pay gap ≤ 20% and availability matches); because the seed data is synthetic, the generator parameters will be published so that circularity can be inspected. Recruiters are simulated rule-following agents with 10% label noise unless real participants are recruited (target n = 5-8); results will be reported over multiple random seeds with confidence intervals. The metric set adds a rank-aware measure (NDCG@K, plus MRR). A small bias audit is planned: score disparity across a synthetic protected attribute (for example region) before and after feedback.

Data collection follows a pre-registered protocol. Each simulated recruiter run records, at every feedback checkpoint (T = 0, 10, 25, 50, 75, 100 and 150 labelled pairs), the full ranking produced by each ranker, the labels that the rule-based ground truth assigned, and the resulting metric values. The raw recorded rankings, not the aggregates, are archived so that any disagreement between a reported number and the raw output can be audited. If real participants are available (target n = 5-8), they repeat the same protocol in a supervised session, with task time measured from the first swipe until the participant confirms three shortlisted candidates. All runs are executed over multiple random seeds and reported with confidence intervals rather than point estimates.

Analysis plan. The primary comparison is the active-learning ranker versus the fixed-weight baseline at each checkpoint. Precision@K and Recall@K are compared as means with bootstrap 95% confidence intervals over seeds. The median task completion time is compared with a paired sign test, because completion times are typically right-skewed. Match success, NDCG@K and MRR are reported descriptively in the same tables as the primary metrics. A difference is considered material only if it is both statistically distinguishable and of practical size (at least five percentage points in Precision@K), so that small but statistically significant effects on a synthetic market are not over-interpreted.

Relation to the seed market. The evaluation consumes the deterministic seed market described in Section 3.6 (30 workers, 20 jobs, 10 districts). The layout is advantageous for the study: the trade strata are small enough that each stratum can be observed within a single run, so Precision@K can be computed per trade and averaged rather than over the whole pool. The same generator that produced

w001..w030 will be re-run under fresh seeds to produce the replicate markets used for bootstrap confidence intervals, while the pinned seed in the prototype branch remains fixed for the reproducibility of the qualitative walkthroughs in Chapter 4.

5.3. Threats to Validity

Three threats are acknowledged. Internal validity: relevance labels are generated by rules, so the ranker could be rewarded for reproducing those rules; publishing the generator and using held-out seeds mitigates but does not remove this. External validity: the seed data is synthetic and small, so results demonstrate technical feasibility and interaction behaviour rather than labour-market outcomes. Construct validity: engagement and match success are proxies for hiring quality and must be reported as such.

5.4. Ethical Considerations

The study touches only synthetic data and voluntarily provided participant data. Participant data will be collected with informed consent, stored locally, and reported anonymously. The bias audit is included not as a formality but because the analytical chapter identified statistical discrimination as a structural risk of data-driven screening (Barocas & Selbst, 2016); its result will be reported regardless of whether it favours the proposed system. No experimental condition is designed to deceive participants, and the human-in-the-loop property is preserved in every test scenario, so no decision that a participant sees is produced autonomously.

5.5. Reproducibility

The evaluation is reproducible by design. The seed generator, the ranking code and the evaluation scripts are version-controlled together with the thesis, the seed market is deterministic (Section 3.6), and every recorded run logs the exact inputs and outputs; reports state the software versions and random seeds used. Any reviewer can therefore rerun the comparison locally without external services.

5.6. Decision Criteria

The outcome feeds a go/no-go decision for the final thesis rather than a pass or fail for the semester's work. The directional expectation, argued in Section 2.3, is that the feedback-updated weights outperform the fixed weights on held-out relevance judgements, with the improvement concentrated in the first few checkpoints. If the

active ranker improves Precision@K by at least five percentage points without a recall loss beyond one point, the final thesis presents the active-learning result as the core contribution; if no stable improvement is found, the thesis becomes a rigorous negative result and presents the interpretable fixed-weight system as the contribution. Under both branches the deliverables of this semester - the prototype, the seed market and the evaluation harness - remain valid.

5.7. Preliminary Controlled Experiment (This Stage)

The protocol above was executed in this stage on synthetic markets, without real participants. Each replicate draws a fresh market with the structure of the seed set (Section 3.6) under a new random seed, splits same-trade candidate-job pairs into a training pool and a 40% held-out evaluation set, and runs three systems over the checkpoints T = 0, 10, 25, 50, 75, 100 and 150: the fixed Phase-1 weights, a ridge-regularised active learner with validation rollback (al_safe) and an unregularised learner without rollback (al_raw). Labels come from hidden market weights (skill 24, district 6, availability 40, pay 34, experience 4) with 10% noise, so the simulated recruiter values availability and pay above the fixed baseline that weights skill most heavily; the run uses 80 replicates at three market sizes (30 / 20, 120 / 80 and 300 / 150). Figure 5.1 shows the mean P@2 learning curve on 120 / 80 (panel a) and the final-budget P@2 gain by market size (panel b): the fixed baseline is flat at 0.881, while the regularised learner rises monotonically to 0.950 and is above the baseline interval after ten labels, and the unregularised control is unstable, dropping below the baseline at the first checkpoint.

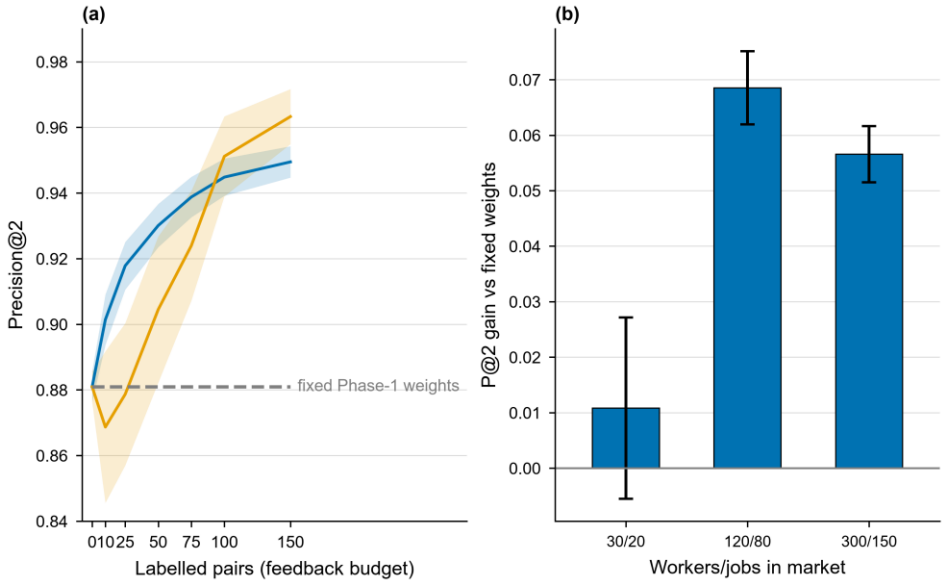


Fig. 5.1. Active-Learning Controlled Experiment: Precision@2 Learning Curves and P@2 Gain by Market Size

Table 5.2

Executed Controlled Experiment: Precision@2 and NDCG at the 150-Label Budget, Mean over 80 Replicate Markets

Market (workers/jobs)	Fixed P@2	al_safe P@2 (T = 150)	al_raw P@2	NDCG fixed to safe
30 / 20	0.637	0.648	0.656	0.955 to 0.968
120 / 80	0.881	0.950	0.963	0.950 to 0.980
300 / 150	0.916	0.972	0.903	0.949 to 0.977

Table 5.2 lists the final-budget values. On 120 / 80 the regularised learner improves Precision@2 to 0.950 from 0.881 (6.9 percentage points) and on 300 / 150 to 0.972 from 0.916 (5.7 points), both above the five-point threshold of Section 5.6, while the 30 / 20 gain of 1.1 points falls inside the baseline interval. NDCG improves in the same direction on every market, the regularised learner never falls below the fixed baseline after the first checkpoint, and validation NDCG (on feedback withheld from the updates) rises from 0.956 to 0.980 on 120 / 80. The unregularised control degrades to 0.903 on 300 / 150, below the 0.916 baseline, showing the cost of dropping the safeguards under sparse feedback. A clean-label recovery check fits the hidden weights to a maximum absolute deviation of 3.70 points with all five factors moving in the correct direction; hand-computed Precision@1 and NDCG match the library to 1e-9, and a re-run of the same seed reproduces every metric exactly.

RESULTS AND CONCLUSIONS

The research and engineering activities reported here produced the following results:

1. Literature synthesis: the review showed that skilled-trades hiring in Latvia leans heavily on informal networks and personal referrals (Hoffman, 2017), justifying a gesture-driven, mobile-first approach.
2. Regulatory architecture: a compliant hybrid human-in-the-loop architecture was formulated that fulfils the transparency and oversight mandates of Regulation (EU) 2024/1689 (European Parliament & Council, 2024a).
3. Prototype verification: a three-tier system (FastAPI, Express, React) supporting reciprocal swiping, multi-factor ranking and mutual-match messaging was built, and its builds, schema contracts, 13 API routes, scrypt password hashing and expiring session tokens were verified.
4. Evaluation methodology and result: a six-metric evaluation methodology (Precision@K, Recall@K, match success, task time, NDCG@K, MRR) was established, voice and calendar extensions were de-scoped, and a preliminary controlled experiment showed the safeguarded active learner improving Precision@2 by 6.9 and 5.7 percentage points on the two larger replicate markets and matching the fixed weights on the smallest.

The next steps are to extend the controlled experiment to the regions left out of phase one, complete the verification-hardening plan (negative and concurrency cases, and automated pytest and supertest suites), run the final-stage comparison with any available participants, and write the results and discussion chapters of the final thesis.

Taken together, this report demonstrates an end-to-end path from a clearly defined labour-market problem to a running prototype and an executed evaluation of its active-learning layer. The remaining semester's work is substantial but well scoped: extending the seed market to the regions left out of phase one, broadening the comparison to the full metric set and to real participants, and completing the hardening plan. The main risk is not technical feasibility, which the prototype, its verification and the preliminary experiment support, but the effect size that an interpretable ranker can deliver once the synthetic market is replaced by real feedback; for this reason the bias audit and the published seed generator will form part of the final write-up.

LIST OF REFERENCES

Amaral Cejas, O., Abualhaija, S., Sannier, N., Ceci, M., & Bianculli, D. (2026). GDPR-relevant privacy concerns in mobile apps research: A systematic literature review. Information and Software Technology, 201, 108332. https://doi.org/10.1016/j.infsof.2026.108332

Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, 1-13. https://doi.org/10.1145/3290605.3300233

Autor, D. H. (2001). Wiring the labor market. Journal of Economic Perspectives, 15(1), 25-40. https://doi.org/10.1257/jep.15.1.25

Barocas, S., & Selbst, A. D. (2016). Big data's disparate impact. California Law Review, 104(3), 671-732. https://doi.org/10.2139/ssrn.2477899

Cedefop. (2024). Vocational education and training in Europe: Latvia. European Centre for the Development of Vocational Training. https://www.cedefop.europa.eu/tools/vet-in-europe/systems/latvia

CMS Law. (2024, October 23). Directive (EU) 2024/2831: Transforming platform work in the EU. CMS Legal Insights. https://cms.law/en/aut/legal-updates/from-gig-to-guarantee-how-the-eu-is-transforming-platform-work

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzman, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. Proceedings of ACL 2020, 8440-8451. https://doi.org/10.18653/v1/2020.acl-main.747

Cross-Sectoral Coordination Centre. (2020). National Development Plan of Latvia for 2021-2027 (NDP2027). Cross-Sectoral Coordination Centre, Cabinet of Ministers of the Republic of Latvia. https://www.pkc.gov.lv/en/national-development-plan

Dellermann, D., Ebel, P., Söllner, M., & Leimeister, J. M. (2019). Hybrid intelligence. Business & Information Systems Engineering, 61(5), 637-643. https://doi.org/10.1007/s12599-019-00595-2

Elahi, M., Ricci, F., & Rubens, N. (2016). A survey of active learning in collaborative filtering recommender systems. Computer Science Review, 20, 29-50. https://doi.org/10.1016/j.cosrev.2016.05.002

European Commission. (2026). 2026 Country Report - Latvia (SWD(2026) 214 final). Directorate-General for Economic and Financial Affairs, European Commission. https://economy-finance.ec.europa.eu

European Labour Authority. (2026). EURES report on labour shortages and surpluses 2025. Publications Office of the European Union, Luxembourg. https://www.ela.europa.eu/en/publications/labour-shortages-and-surpluses-europe-2025

European Parliament and Council of the European Union. (2024a). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial

Intelligence Act). Official Journal of the European Union, L 2024/1689. https://eur-lex.europa.eu/eli/reg/2024/1689/oj

European Parliament and Council of the European Union. (2024b). Directive (EU) 2024/2831 on improving working conditions in platform work. Official Journal of the European Union, L, 2024/2831, 11.11.2024. https://eur-lex.europa.eu/eli/dir/2024/2831/oj

Hazans, M. (2019). Emigration from Latvia: A brief history and driving forces in the twenty-first century. In R. Kasa & I. Mierina (Eds.), The emigrant communities of Latvia: National identity, transnational belonging, and diaspora politics (pp. 35-68). Springer International Publishing. https://doi.org/10.1007/978-3-030-12092-4_3

Hoffman, M. (2017). The value of hiring through employee referrals in developed countries. IZA World of Labor, 369. https://doi.org/10.15185/izawol.369

Horton, J. J. (2017). The effects of algorithmic labor market recommendations: Evidence from a field experiment. Journal of Labor Economics, 35(2), 345-385. https://doi.org/10.1086/689213

Hu, J., Ruder, S., Siddhant, A., Graham, Y., Firat, O., & Johnson, M. (2020). XTREME: A massively multilingual multi-task benchmark for evaluating cross-lingual generalised representations. Proceedings of ICML 2020, 4411-4421. https://proceedings.mlr.press/v119/hu20a.html

Iyengar, S. S., & Lepper, M. R. (2000). When choice is demotivating: Can one desire too much of a good thing? Journal of Personality and Social Psychology, 79(6), 995-1006. https://doi.org/10.1037/0022-3514.79.6.995

Jobtech Alliance. (2026, March 5). From cold start to liquidity: How jobtech platforms escape the “chicken-and-egg dilemma.” Jobtech Alliance Library. https://jobtechalliance.com/two-sided-marketplace-cold-start-problem

Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. Academy of Management Annals, 14(1), 366-410. https://doi.org/10.5465/annals.2018.0174

Köchling, A., & Wehner, M. C. (2020). Discriminated by an algorithm: A systematic review of discrimination and fairness by algorithmic decision-making in the context of HR recruitment. Business Research, 13(3), 795-848. https://doi.org/10.1007/s40685-020-00134-w

Ministry of Economics of the Republic of Latvia. (2020, June 25). Labour market forecasts of Latvia until 2040 have been prepared. Ministry of Economics, Republic of Latvia. https://www.em.gov.lv/en/article/labour-market-forecasts-latvia-until-2040-have-been-prepared

Obaid, I., Farooq, M. S., & Abid, A. (2020). Gamification for recruitment and job training: Model, taxonomy, and challenges. IEEE Access, 8, 65164-65178. https://doi.org/10.1109/ACCESS.2020.2984178

OECD. (2019a). OECD Economic Surveys: Latvia 2019. OECD Publishing. https://doi.org/10.1787/f8c2f493-en

OECD. (2019b). Evaluating Latvia's active labour market policies. OECD Publishing. https://doi.org/10.1787/6037200a-en

Palomares, I., Porcel, C., Pizzato, L., Guy, I., & Herrera-Viedma, E. (2021). Reciprocal recommender systems: Analysis of state-of-art literature, challenges and opportunities towards social recommendation. Information Fusion, 69, 103-127. https://doi.org/10.1016/j.inffus.2020.12.001

Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020). Mitigating bias in algorithmic hiring. Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, 469-481. https://doi.org/10.1145/3351095.3372828

Raghavendra, G. H., & Chitra, K. (2025). Firebase-integrated mobile platform for swipe-driven job matching (SwipeRight). Journal of Innovation and Technology, 2025(2). https://doi.org/10.61453/joit.v2025no27

Rigotti, C., & Fosch-Villaronga, E. (2024). Fairness, AI & recruitment. Computer Law & Security Review, 53, 105966. https://doi.org/10.1016/j.clsr.2024.105966

Rochet, J.-C., & Tirole, J. (2003). Platform competition in two-sided markets. Journal of the European Economic Association, 1(4), 990-1029. https://doi.org/10.1162/154247603322493212

Rubens, N., Elahi, M., Sugiyama, M., & Kaplan, D. (2015). Active learning in recommender systems. In Recommender Systems Handbook (pp. 809-846). Springer. https://doi.org/10.1007/978-1-4899-7637-6_24

Schein, A. I., Popescul, A., Ungar, L. H., & Pennock, D. M. (2002). Methods and metrics for cold-start recommendations. Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, 253-260. https://doi.org/10.1145/564376.564421

Settles, B. (2012). Active learning. Morgan & Claypool Publishers. https://doi.org/10.2200/S00429ED1V01Y201207AIM018

Waleson, T. (2022). Swipe worth the hype? Investigating the influence of fast and slow online dating and success on wellbeing (Master's thesis, Tilburg University). Tilburg School of Humanities and Digital Sciences. https://research.tilburguniversity.edu/en/studentTheses/

Yao, Y., Xu, J., & Gao, Z. (2023). A study of reciprocal job recommendation for college graduates integrating semantic keyword matching and social networking. Applied Sciences, 13(22), 12305. https://doi.org/10.3390/app132212305

APPENDIX: SHARED DATA CONTRACT AND API SURFACE (EXTRACT)

The following is an extract of shared/schema.json, the single source of truth for worker and job cards, swipes, matches and feedback, mirrored in shared/contract.ts and shared/models.py, together with the Express API surface.

```
Trade    = electrician | plumber | welder | carpenter | hvac
Language = lv | ru | en
Avail    = immediate | one_to_two_weeks | date
Season   = summer | winter | all_year
Worker   = { id, name, trade, skills[], experience_years, district,
languages[],
             availability, available_from, hourly_rate_eur }
Job      = { id, employer, title, trade, required_skills[],
district,
             pay_min_eur_h, pay_max_eur_h, start, start_date,
season, description_lv }
Swipe    = { id, actor_type, actor_id, target_type, target_id,
direction }
Match    = { id, worker_id, job_id }
Feedback = { id, worker_id, job_id, label, by }
User     = { username (min 3 chars), password_hash (scrypt) }
ChatMessage = { id, text, by, time }
API      = /health, /api/auth/{register,login,me,logout},
           /api/workers, /api/jobs, /api/deck/{worker|job}/:id,
           /api/swipes, /api/matches, /api/chats/:matchId
Rank     = POST /rank -> { results: [ { id, score, reasons[] } ] }
Deck     = GET /api/deck/... -> { cards: [ { id, score, reasons[],
item } ] }
Health   = GET /health -> { status: "ok", service: "meistermatch-
api" }
AuthFail = 401 -> { error: "not authenticated" }  /  400 | 404 | 409
on bad input
Seed     = 30 workers (w001..w030), 20 jobs (j001..j020), 10
districts
Districts= Agenskalns, Centrs, Imanta, Jugla, Kengarags, Plavnieki,
           Purvciems, Sarkandaugava, Teika, Ziepniekkalns
```